■■ Drone Intelligence System for India

End-to-End Production AI/ML Architecture & Assessment Documentation

Project Overview: This system serves as India's comprehensive drone knowledge hub, featuring a **Multi-Query RAG Pipeline** with persistent **ChromaDB**, a **Model Context Protocol (MCP) Calculation Server**, a **FastAPI REST Backend**, and an interactive **React 19 + TypeScript Dashboard**.

Problem Solved: Consolidates DGCA Drone Rules 2021 (amended 2023), DigitalSky airspace zones, Namo Drone Didi subsidies, PLI schemes, and Indian OEM specifications into a single intelligent platform.

■ End-to-End System Architecture Flowchart

React 19 Dashboard Real-time Chat, Calculators, Analytics	■	**FastAPI REST Backend** Intelligent 95% Tool Router
LangGraph RAG Engine Multi-Query + ChromaDB + RRF	■	**MCP Calculation Server** Flight, ROI, Compliance, Specs

■ Phase-by-Phase Technical Implementation

Phase 1: Research & Data Collection

Compiled authoritative data on Indian drone OEMs (ideaForge, Marut, IoTechWorld, Garuda), DGCA Drone Rules 2021, DigitalSky zones, Namo Drone Didi subsidies, and PLI schemes into handbooks in data/raw/.

Phase 2: Data Generation & Dataset Creation

Created structured catalogs (drone_models.json, rules_regulations.json), markdown handbooks, and synthetic scripts (generate_synthetic_data.py) generating 500+ flight telemetry records and 100 farm ROI scenarios.

Phase 3: RAG System Implementation

Built a LangGraph StateGraph pipeline with persistent ChromaDB vector store, Multi-Query expansion, Hybrid BM25 + Vector Search, RRF Reranking with keyword boost, and Gemini LLM synthesis with explicit source citations.

Phase 4: MCP Server Development

Developed a standalone Model Context Protocol server exposing 4 calculation tools: Flight Time Calculator, Agriculture ROI Calculator, DGCA Compliance Checker, and Drone Selection Assistant.

Phase 5: FastAPI Backend Development

Created modular routes (api/routes/) supporting /chat, /upload, /calculate/*, /check/compliance, /recommend/drone, /analytics, and persistent SQLite session history.

Phase 6: Interactive React Dashboard

Built a responsive React 19 + TypeScript frontend featuring AI Chat agent, interactive Recharts visualizations, drag-and-drop document upload, dark/light mode toggle, and result export.

■ API Endpoint Reference

Endpoint Route	HTTP Method	Description & Payload
/api/chat	POST	Hybrid RAG + MCP Query with persistent session history
/api/upload	POST	Uploads PDF/MD document, extracts text & tables, seeds ChromaDB
/api/calculate/flight-time	POST	Flight duration, range, battery drawdown curve, weather advice
/api/calculate/roi	POST	Agriculture ROI, payback period, 3-year profit projection
/api/check/compliance	POST	DGCA Airspace compliance status (Green/Yellow/Red zones)
/api/recommend/drone	POST	Ranks Indian drone models by budget, payload, & specs
/api/analytics	GET	System usage statistics, P50/P95 latency, popular queries
/api/chat/history/{session_id}	GET / DELETE	Retrieves or clears persistent chat history for a session

■ Automated Testing & Verification

Pytest Test Suite Status: 18 Passed, 0 Failed (Execution Time: 2.46s)
• Coverage includes: API Health Check, RAG Chat Endpoint, Flight Time Calculator, ROI Engine, DGCA Compliance Checker, Drone Recommender, Session History Persistence, Document Ingestion Upload, and RRF Reranker.

■ Docker & Quick Start Execution

```
# 1. Run Data Generation & Seed ChromaDB Vector Database
python scripts/generate_synthetic_data.py
python scripts/preprocess_data.py
python scripts/seed_vector_db.py

# 2. Launch FastAPI Backend Server (:8000)
python -m uvicorn api.main:app --reload --port 8000

# 3. Run Containerized System via Docker Compose
docker-compose up --build
```

■ Submission Guidelines & Contact Information

Assessment Submission: JulleyOnline AI/ML Internship Project (Round 1)
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Repository: Public GitHub Repository with clean commit history & CI/CD workflow.